

Consumer Shopping Behavior Analysis Project

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

The objective of this project is to analyze the company's consumer behavior dataset and answer the following overarching business question:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Project Objectives

The main objectives of this project are:

- Analyze customer purchasing behavior across different demographics and sales channels.
- Identify factors influencing buying decisions and repeat purchases.
- Discover trends related to product categories, discounts, seasons, and payment methods.
- Build interactive dashboards for business decision-making.
- Provide actionable recommendations to improve customer engagement and marketing strategies.

Deliverables

1. Data Preparation & Modeling (Python)

- Clean and preprocess the raw dataset.
- Handle missing values, duplicates, and inconsistencies.
- Perform data transformation and feature engineering.
- Prepare the dataset for SQL analysis and dashboard visualization.

2. Data Analysis (SQL)

- Design structured database tables for business transactions.
- Write SQL queries to analyze:
 - Customer segments
 - Purchase frequency
 - Loyalty behavior
 - Product performance
 - Sales trends
 - Discount effectiveness

- Extract insights related to consumer purchase drivers.

3. Visualization & Insights (Power BI)

- Build an interactive Power BI dashboard.
- Visualize:
 - Sales performance
 - Customer demographics
 - Seasonal trends
 - Online vs. offline sales
 - Payment preferences
 - Repeat purchase behavior
- Enable stakeholders to make data-driven decisions through interactive reports.

4. Report and Presentation

- Prepare a detailed project report summarizing:
 - Methodology
 - Analysis process
 - Key findings
 - Business insights
 - Recommendations
- Create a professional presentation with visual insights and actionable strategies for stakeholders.

5. GitHub Repository

- Maintain a well-structured GitHub repository containing:
 - Python scripts
 - SQL queries
 - Power BI dashboard files
 - Documentation and README

Expected Outcomes

By completing this project, the company will be able to:

- Understand customer purchasing behavior more effectively.
- Improve targeted marketing strategies.
- Increase customer retention and loyalty.
- Optimize product offerings and discount strategies.
- Enhance overall business performance using data-driven insights.